NetFRAME

Network Topology & Systems Reference

`192.168.10.0/24 · km-cluster · 7 nodes · 3 GPUs · 4× RAIDZ2`

Owner	Kyle Mason (`machismo0311`) — Vermilion / Greater Cleveland, OH
Document	NetFRAME Network Topology & Systems Reference
Version	1.0
Generated	2026-07-08
Classification	Public / Sanitized (service tags & serials omitted)
Method	Live discovery (nmap · Proxmox API · SSH facts · Headscale · ZFS) reconciled against the Home-Lab vault
Scope	Every host, VLAN, subnet, link, service, pool, and overlay path

Single-site production-grade home lab · Proxmox VE 9.2.3 · Juniper-cored VLAN fabric · ZFS + PBS · private GPU LLM platform

Contents

1 Executive Summary

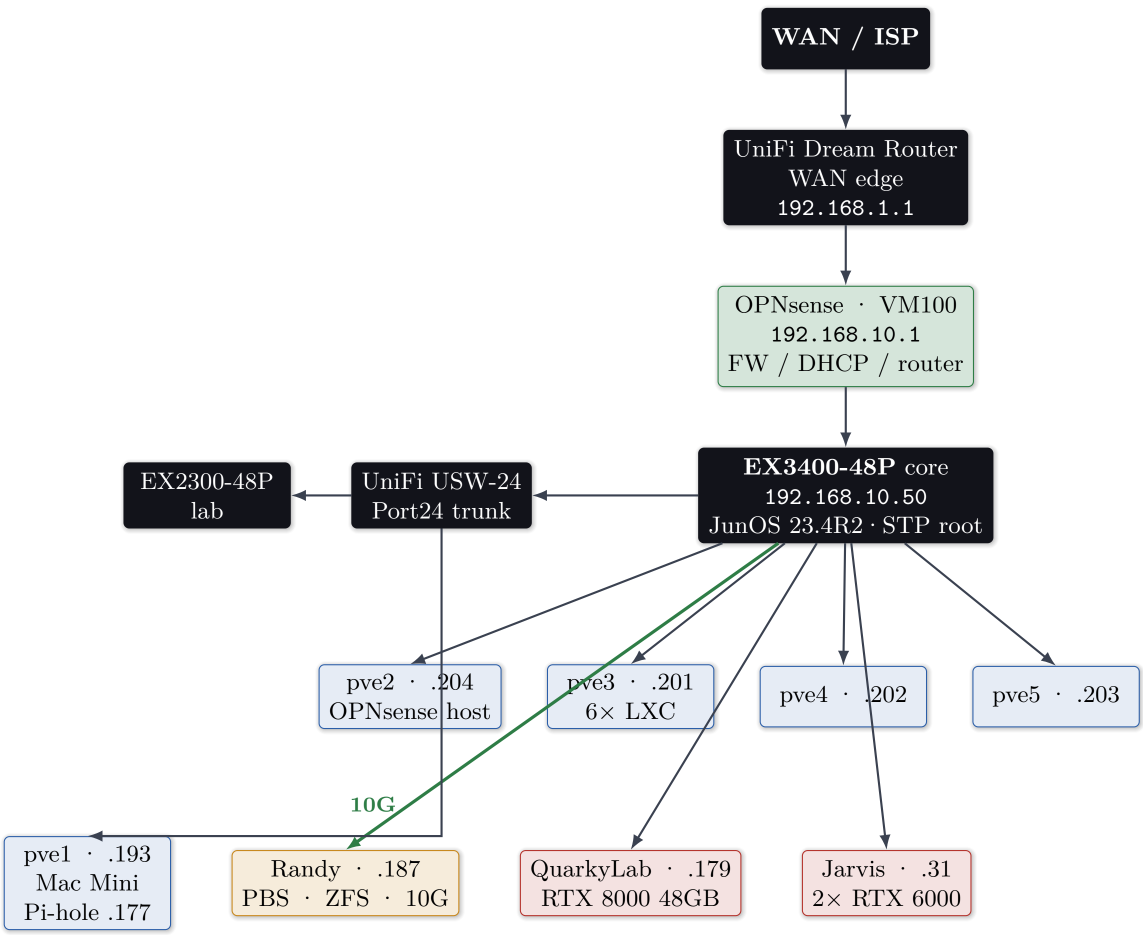
NetFRAME is a single-site, 42U home lab built around a **7-node Proxmox VE 9.2.3 cluster** (`km-cluster`), a **Juniper-cored, VLAN-segmented fabric**, and a **24-bay ZFS storage node** providing NFS and Proxmox Backup Server. It runs reverse-proxied web services with automated TLS, a Prometheus/Grafana/Loki observability stack, drive-health telemetry across ~50 disks, a Wazuh SIEM, an internal ACME PKI, a password vault, and a **GPU-accelerated private LLM platform** (RTX 8000 + dual RTX 6000, ~96 GB VRAM) with a custom OpenAI-compatible router that transparently escalates to the Claude API.

Proxmox nodes (quorate)	7/7	ZFS raw / usable	36.7 TB / ~23 TB
Standalone hypervisor	1 (pve1)	PBS datastore	24.6 TB
VMs / LXCs	2 / 6	Monitored drives	~50
Managed switches	3	Overlay VPN nodes	9
VLANs defined	7	Aggregate RAM	~1.5 TB
Physical GPUs / VRAM	3 / ~96 GB	UPS (split-bus)	2 / ~2220 W

Edge model — deliberate double-NAT

A UniFi Dream Router is the WAN edge (`192.168.1.0/24`); **OPNsense** (VM 100 on `pve2`) is the LAN router / firewall / DHCP for `192.168.10.0/24` and the inter-VLAN gateway. The lab LAN therefore sits behind its own firewall boundary.

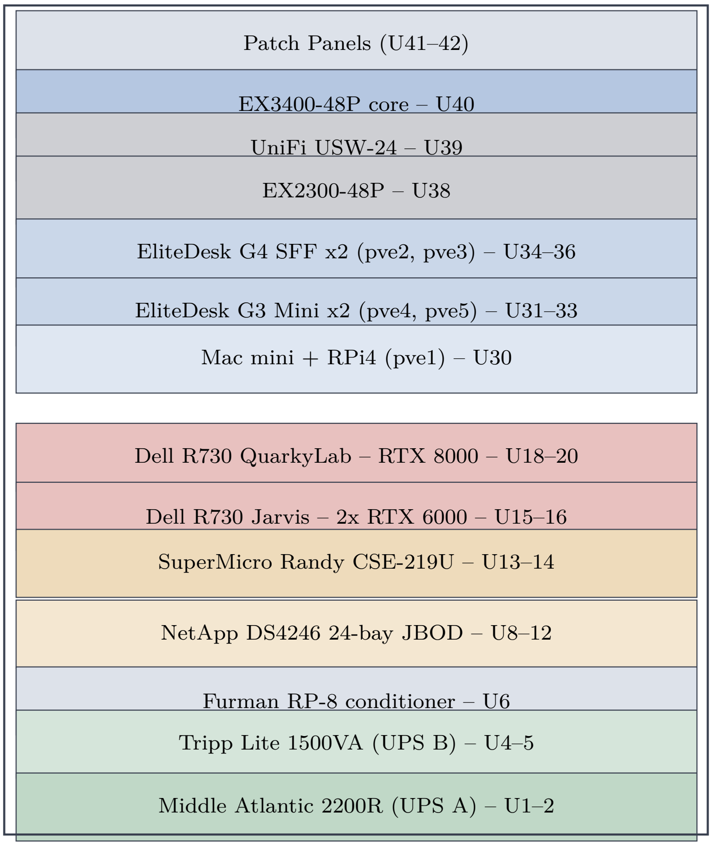
2 Physical & Logical Topology



Thick green = 10 GbE (Randy `xe-0/2/0`; Jarvis `xe-0/2/2`). All other links 1 GbE.

3 Layer 1 — Physical

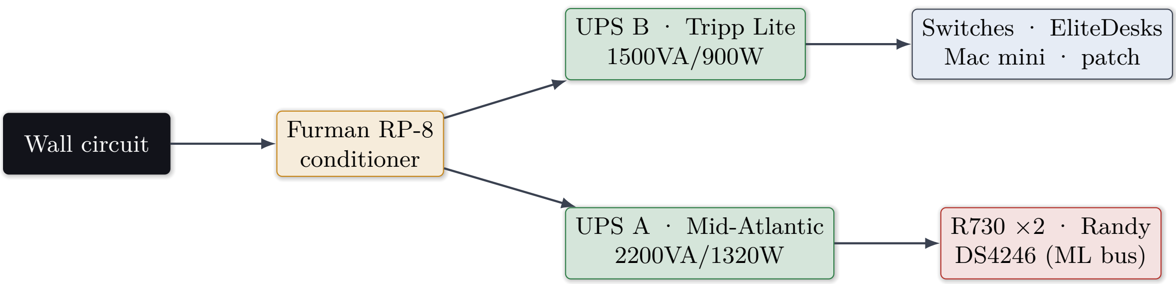
3.1 Rack elevation (NetFRAME CS9000, 42U — rear panel removed for R730 depth)



3.2 Key physical links

From	Port	To	Speed	Notes
EX3400	`ge-0/0/46`	UniFi USW-24 P24	1G	802.1Q trunk (native 1 + tagged 20/30/40/50/60/70)
EX3400	`ge-0/0/45`	EX2300	1G	Trunk, all VLANs
EX3400	`xe-0/2/0`	Randy nic3 (Mellanox)	**10G**	Storage / NFS data path
EX3400	`xe-0/2/2`	Jarvis enp132s0 (ConnectX)	**10G**	Access VLAN 30
EX3400	`ge-0/0/24`	QuarkyLab nic2	1G	Trunk native 1 + tagged 30
EX3400	`ge-0/0/22`	Jarvis nic2	1G	Native VLAN 1 (mgmt/corosync)
EX3400	`ge-0/0/30·32·44`	iDRAC/IPMI OOB	1G	Tagged VLAN 20 only
EX3400	`xe-0/2/3`	UniFi SFP2 DAC	—	**DOWN** — 10G/1G EEPROM mismatch

3.3 Power — split-bus one-line



Capacity watch

UPS A carries both GPU R730s. RTX 8000 alone can draw ~260 W under CUDA; with Jarvis's 2× RTX 6000 online, full ML load approaches the 1320 W continuous rating. Both UPS units are exposed via **NUT 2.8.1** on `pve3:3493` (UPS A over SNMP card `192.168.10.180`, UPS B over USB).

4 Layer 2 — Switching & VLANs

4.1 Switch fabric

Device	Mgmt IP	OS	Role
EX3400-48P	`192.168.10.50`	JunOS 23.4R2-S7.4	**Core** — 48×1G PoE+, 4×10G SFP+, 2×40G QSFP+, dual PSU, STP root (prio 4096)
UniFi USW-24-250W	UniFi-managed	UniFi	Access / AP aggregation; Port 24 trunk
EX2300-48P	via trunk	JunOS	Secondary / lab isolation
UniFi Dream Router	`192.168.1.1`	UniFi	WAN edge (upstream of OPNsense)

4.2 VLAN matrix

VLAN 1	`192.168.10.0/24`	Management
VLAN 20	`192.168.20.0/24`	Trusted/OOB
VLAN 30	`192.168.30.0/24`	Servers
VLAN 40	`192.168.40.0/24`	IoT
VLAN 50	`192.168.50.0/24`	VoIP
VLAN 60	`192.168.60.0/24`	Guest
VLAN 70	`192.168.70.0/24`	Lab

Gateways are OPNsense SVIs (`192.168.X.1`); EX3400 carries the L3 mgmt SVI on `irb.10`.

JunOS ELS gotcha (live)

On this EX3400, `native-vlan-id` must be set at the **physical-interface** level, *not* under `unit 0 family ethernet-switching`. Misplacement previously caused a trunk outage. The tagged trunk to UniFi is `ge-0/0/46`.

5 Layer 3 — Addressing, Routing, DNS

5.1 Static IP registry (VLAN 1)

IP	Host	Role
`.1`	OPNsense	LAN gateway / FW / DHCP (VM100, pve2)
`.2`	UniFi USW-24	Access switch mgmt
`.31`	Jarvis	R730 LLM node
`.50`	EX3400	Core switch
`.100/.199`	Ares	Admin workstation (wired / WiFi)

IP	Host	Role
`.148`	Homepage	LXC 106
`.177`	Pi-hole	LXC 103 on pve1 (Mac Mini)
`.179`	QuarkyLab	R730 ML node
`.180`	UPS A SNMP	Mid-Atlantic OL2200R card
`.181`	Nginx Proxy Mgr	LXC 101
`.182`	Vaultwarden	LXC 102
`.183`	Grafana/Prom/Loki/Influx/Scrutiny	LXC 103
`.184`	Wazuh SIEM	VM 104 (QuarkyLab)
`.185`	Open WebUI	LXC 107
`.186`	Headscale	LXC 105
`.187`	Randy	PBS / ZFS / Jellyfin
`.193`	pve1	Mac Mini standalone (Pi-hole host)
`.201-.204`	pve3/pve4/pve5/pve2	Cluster nodes

5.2 Routing — per-node default gateway (live)

Node	Default GW	Iface	Note
pve2	`192.168.10.1`	vmbr1	correct (OPNsense)
pve3/4/5	`192.168.1.1`	vmbr0	**onlink to UDR — off-subnet** (finding F-1)
QuarkyLab	`192.168.30.1`	vmbr0.30	VLAN 30 egress
Jarvis	`192.168.30.1`	vmbr1	VLAN 30 egress (10G)
Randy	`192.168.30.1`	vmbr0.30	VLAN 30 egress

5.3 DNS & OOB / Servers VLANs

Pi-hole (`192.168.10.177`, FTL v6, LXC on pve1) is the LAN resolver + ad/tracker sink and serves `*.netframe.local` (e.g. `llm./chat.` → NPM `.181`). Public `*.kylemason.org` resolves via Cloudflare (grey-cloud) to NPM.

IP	Host	Note
`192.168.20.20/21/22`	QuarkyLab iDRAC / Jarvis iDRAC / Randy IPMI	OOB VLAN 20 (2026-07-03); credentials rotated
`192.168.30.179/187/31`	QuarkyLab / Randy / Jarvis	NFS / PBS / egress (dual-homed)

6 Overlay — Headscale Mesh VPN

Self-hosted Headscale v0.29.1 (LXC 105, `192.168.10.186:8080`), tailnet `100.64.0.0/10`.

Tailnet IP	Node	State	
`100.64.0.1` Ares (offline)	`100.64.0.2` Randy	`100.64.0.3` pve5	`100.64.0.4` pve4
`100.64.0.5` pve3	`100.64.0.6` Jarvis	`100.64.0.7` QuarkyLab	`100.64.0.8` pve1
`100.64.0.9` Fernanda-Mac	tag:ssh; all nodes online except laptops		

7 Compute & Virtualization — km-cluster

7.1 Node hardware inventory

Node	Chassis	CPU	RAM	GPU	Kernel
pve2	EliteDesk 800 G4	i7-8700	31G	—	7.0.12-pve
pve3	EliteDesk 800 G4	i7-8700	46G	—	7.0.12-pve
pve4	EliteDesk 800 G3 Mini	i5-7500T	31G	—	7.0.12-pve
pve5	EliteDesk 800 G3 Mini	i5-7500T	31G	—	7.0.12-pve
QuarkyLab	Dell R730	2× E5-2699v4	503G	RTX 8000 48G	**6.14.11-pve**
Jarvis	Dell R730	2× E5-2687Wv4	377G	2× RTX 6000	**6.14.11-pve**
Randy	SuperMicro CSE-219U	2× E5-2690v4	125G	RX 580	7.0.12-pve
pve1	Mac Mini (2011)	—	—	—	standalone

Kernel pinning — do not upgrade

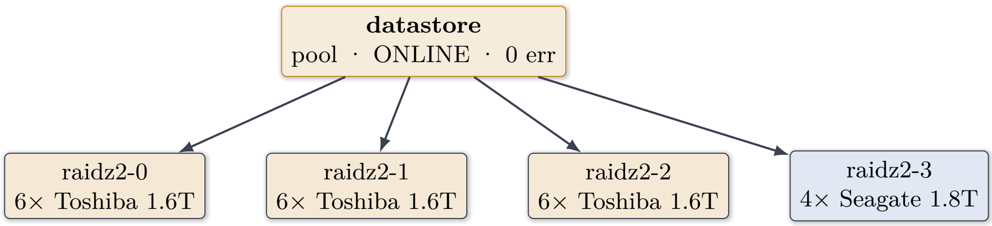
QuarkyLab and Jarvis are pinned to **6.14.11-9-pve** via `GRUB_DEFAULT`; 6.17+ breaks the NVIDIA 550.163.01 stack. Never run kernel upgrades or change GRUB default on either GPU node.

7.2 VM / LXC inventory

ID	Name	Host	Res	IP	Payload
VM100	OPNsense	pve2	4G	`.1`	LAN router/FW/DHCP v25.7
VM104	Wazuh	QuarkyLab	16G	`.184`	SIEM (no guest-agent — see §Security)
101	nginx-proxy	pve3	512M/8G	`.181`	Nginx Proxy Manager
102	vaultwarden	pve3	512M/10G	`.182`	Vaultwarden
103	grafana	pve3	2G/20G	`.183`	Grafana+Prom+Loki+Influx+Scrutiny
105	headscale	pve3	512M/4G	`.186`	Mesh VPN control
106	homepage	pve3	512M/8G	`.148`	Homepage + PeaNUT
107	openwebui	pve3	4G/16G	`.185`	Open WebUI

8 Storage Architecture

8.1 Randy ZFS `datastore` — 4× RAIDZ2 (36.7 TB raw / ~23 TB usable)



Boot: RAID-1 mirror (2× ST200FM0053 SAS) on **AVAGO 3108 MegaRAID**; data drives on a **separate LSI 9207-8e in IT mode**. PBS v4.2.2 (`192.168.10.187:8007`) exposes a 24.6 TB datastore.

Live finding — DS4246 dual-path SAS

The NetApp DS4246 shelf's 4 TB drives enumerate as **duplicate serial pairs** — dual-path SAS multipath through the shelf's two I/O modules presenting each disk on two paths. Configure `multipath`/dm before pooling. (This also flags drift: docs list 2 TB drives; live shelf is 4 TB-class, buildout in progress.)

PBS storage pin (F-4, resolved)

The VLAN-30 migration silently repointed PBS storage to `.30.187` and stalled VLAN-1 node backups from 2026-07-02; fixed 2026-07-06 back to `192.168.10.187`. **Keep PBS storage on `.10.187`.**

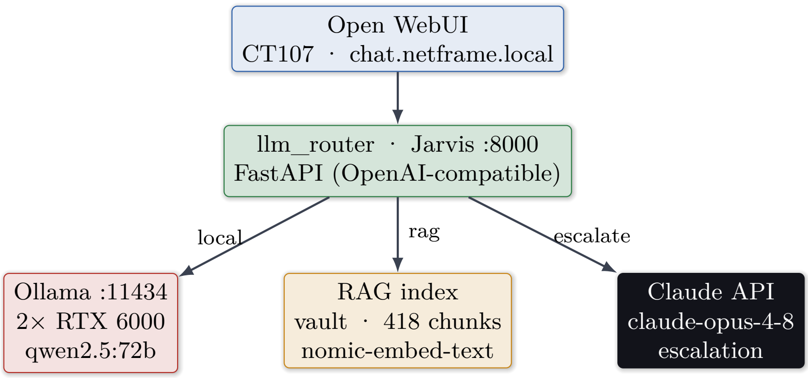
9 Services & Application Layer

9.1 Reverse proxy & published services (NPM, CT101)

Hostname	Backend	TLS	Auth
vault.kylemason.org	`.182`	CF DNS-01	Vaultwarden
grafana.kylemason.org	`.183:3000`	CF DNS-01	Grafana
homepage.kylemason.org	`.148:3000`	CF DNS-01	Basic (kyle)
wazuh.kylemason.org	`.184:443`	CF DNS-01	Wazuh
chat.netframe.local	Open WebUI `.185`	internal	admin
llm.netframe.local	Jarvis `.31:8000`	HTTP	Pi-hole-scoped

Observability: Prometheus (`.183:9090`, localhost-only, 8 exporters), Grafana v13, Loki, InfluxDB, Scrutiny (~50 drives, collectors on Randy/QuarkyLab/Jarvis), NUT (`pve3:3493`), CrowdSec IPS, step-ca PKI (`pve2:443`).

10 AI / LLM Platform



On-prem, cited answers over your own notes

The `"rag"` model grounds answers on the Home-Lab Obsidian vault (418 chunks, cosine index) with `[source]` citations — served entirely on GPU on Jarvis, with a one-flag escalation to Claude Opus for hard queries. QuarkyLab's RTX 8000 48 GB drives a parallel DUNE research RAG agent (Fernanda).

11 Security Architecture

- **OOB isolation (Phase 1):** all three BMCs moved off flat VLAN 1 onto tagged VLAN 20; default credentials rotated to Vaultwarden; reachable only from Ares `enp0s31f6.20`.
- **Servers VLAN 30:** GPU/storage nodes dual-homed (NFS/PBS/egress on 30; corosync/mgmt on 1).
- **Ingress chokepoint:** all web traffic via NPM; admin planes localhost- or Ares-bound (pentest F-03/F-05, verified).
- **Defense in depth:** Wazuh SIEM + CrowdSec IPS + step-ca PKI + Vaultwarden secrets.

Operational — Wazuh guest agent (F-5)

VM 104 lacks `qemu-guest-agent`; a QuarkyLab host reboot hard-stops it and `wazuh-indexer` returns unhealthy. Recovery: `qm stop 104 && qm start 104`, wait ~4 min (healthy = dashboard 302→/login).

12 Data-Flow Walkthroughs

A · Grounded chat: Browser → Pi-hole resolves `chat.netframe.local` → NPM `.181` → Open WebUI `.185` → `llm_router` Jarvis `.31:8000` `(rag)` → embed + cosine search vault → Ollama `qwen2.5:72b` tensor-split → cited answer.

B · Nightly backup: `pvescheduler` → `vzdump` (02:00 CT / 03:00 VM) → PBS `192.168.10.187:8007` (VLAN 1) → prune 7d+4w.

C · Storage: GPU nodes mount Randy NFS over **VLAN 30 10G** (`.30.187`, `xe-0/2/0`); corosync stays on VLAN 1.

D · BMC access: Ares tags VLAN 20 (`enp0s31f6.20`) → EX3400 `ge-0/0/30/32/44` → iDRAC `192.168.20.20-22`.

13 Observations, Drift & Recommendations

#	Sev	Finding	Recommendation
F-1	Med	Inconsistent default GWs: pve3/4/5 via `192.168.1.1` (off-subnet, bypasses OPNsense)	Repoint to `192.168.10.1` or document intent
F-2	Low	Ares wired mgmt leg link-down; VLAN 20 reroutes via WiFi	Restore wired leg; verify `ip route get`
F-3	Med	DS4246 drift + dual-path duplicates	Configure multipath before pooling; update docs
F-4	✓	PBS storage was repointed to `.30.187` (broke backups)	Fixed; keep on `.10.187`; monitor backup age
F-5	Low	Wazuh VM lacks guest-agent	Install `qemu-guest-agent`; cold start
F-6	Low	`xe-0/2/3` DAC down (EEPROM mismatch)	Replace SFP or decommission
F-7	Info	VLANs 40/50/60/70 lightly populated	Populate or prune

14 Failure-Domain / Blast-Radius

Component	If it fails	Mitigation
pve2 (OPNsense)	No LAN routing/DHCP/inter-VLAN — whole LAN	onboot; serial-console runbook
pve3 (6 LXCs)	NPM/Vault/Grafana/Homepage/Headscale/OpenWebUI down	PBS backups; migratable CTs
Randy	Backups + GPU NFS + Jellyfin	RAIDZ2 (2-disk/vdev); split HBAs
EX3400	Entire wired fabric	Dual PSU; UPS B
UPS A	Both R730s + Randy + DS4246	2200VA headroom; graceful shutdown
pve1	LAN DNS / ad-block	Secondary resolver `1.1.1.1`
Corosync	Loss of quorum	7 votes on stable VLAN-1 L2

15 Appendix — Live Port / Service Scan

Verified 2026-07-08 (`nmap -sT -sV`, top-200). `:9100`=node-exporter, `:3128`=Proxmox `pveproxy`. Proxmox `:8006`, PBS `:8007`, Jellyfin `:8096` are outside top-200 but known-live.

Host	Open ports	Identity
`.1` OPNsense	53,80,443	Unbound DNS 1.23.1 + GUI
`.2` UniFi USW	53,80,443,8080,8443	dnsmasq + UniFi
`.31` Jarvis	22,111,3128,**8000**,9100	SSH · Proxmox · llm_router · exporter
`.50` EX3400	22,830	SSH + NETCONF
`.100/.199` Ares	21,22,111,139,445,2049,9090	FTP · Samba · NFS · Cockpit
`.112` Amazon IoT	**1080**,8888	open no-auth SOCKS5 (F-8)
`.148` Homepage	22,8081	SSH · PeaNUT
`.177` Pi-hole	22,53,80,443	dnsmasq pi-hole v2.92.2
`.179` QuarkyLab	22,111,3128,9100	SSH · Proxmox · exporter
`.180` UPS card	22,80,443	CyberPower
`.181` NPM	22,80,81,443	OpenResty proxy
`.182` Vaultwarden	22,80	Vaultwarden
`.183` Observability	22,3000,8080,9100	Grafana · Scrutiny · exporter
`.184` Wazuh	22,443	SIEM dashboard
`.185` Open WebUI	22,8080	Chat UI
`.186` Headscale	22,8080	Mesh VPN control
`.187` Randy	22,111,2049,3128,9100	SSH · NFS · Proxmox · exporter (+8007/8096)
`.201-.204` pve3/4/5/2	22,111,3128,9100	nodes (.204 also 443=step-ca)

Generated 2026-07-08 from live discovery reconciled against the NetFRAME Home-Lab vault.
Sanitized public edition. Sanitized edition suitable for the public repo generated alongside.